

Project Stochastic Simulation

Simulation and analysis of policies for patient scheduling for CT scanners

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1 Introduction

Efficient use of CT scanners is a key challenge for hospitals. Especially in the CT scan department it is really important to have an efficient policy to schedule patients, so that emergency patients get the urgent care they need, but other patients also get their care in reasonable time. Not to forget the financial aspect of a scanner not being used for a time, but still making costs. Having an inefficient way of scheduling patients may lead to excessive waiting times, under utilization of these expensive scanners or delayed care for patients that are critically ill. In this report we provide a simulation model to analyze the current policy a CT scan department is using. This policy distinguishes three types of patients: emergency patients, inpatients, and outpatients and uses two scanners during office hours, and only one outside office hours. In this report we will first analyze whether the capacity of this department is sufficient. This we will do in an analytical way, without the simulation. After that, we will construct and use a simulation model to evaluate the performance of the current policy. The overview of this report is as follows: after this introduction we give a detailed problem description in section 2, in section 3 we will do the analytical capacity check and we set up the simulation model using events, in section 4 we focus on the statistical analysis together with 95% confidence intervals, in section 5 we will show the results, and lastly in section 6 we give a short conclusion.

2 Problem Description in Words

The CT scan department helps three types of patients: emergency patients, inpatients, and outpatients. Each type of patient has their own urgency levels, arrival process, and scheduling restrictions. To scan these patients, the department uses two identical CT scanners. During office hours (08:00-16:00) both of these scanners are turned on. Outside office hours only one of these scanners is available for emergency patients and inpatients. Outpatients only come on appointment. Emergency patients are critical patients which are prioritized when they arrive. Inpatients are patients that are in the hospital and need to be scanned. Inpatients are preferably scanned within office hours. Upon arrival, patients wait in a shared waiting room with limited capacity. In this case, that is three chairs. If additional patients arrive, they then have to wait outside the waiting room, which is not preferred. As said, emergency patients have priority, which is non-preemptive. Other patients wait in a FIFO queue. Outpatients are being scheduled for their arrival, with the restrictions that they can only be scheduled as early as the day after their call and there are only 4 slots available in the morning and 3 in the afternoon. If there is no slot available, then this patient will be put on a waiting list to be scanned in the next week.

To evaluate the current policy we use the following performance measures: CT scanner utilization, access times for outpatients, waiting time for both emergency and outpatients, fraction of patients who have to wait outside the waiting room, and the percentage of inpatients of which the request arrived during office hours that is actually scanned during that same day within office hours.

3 Simulation Model

3.1 Analytical capacity check

In this section we want to explore whether two CT scanners are sufficient to be able to treat all patients. To do so, we will consider the long run workload of the scanners. It can be that in a week the workload is higher, but in the long term we expect that the scan duration converges to its expectation and the number of patients converges to its rate.

For all patients the scan duration is uniform distributed on [10, 19] minutes. That means the expected time of a scan in the long term is

$$E[\text{scan}] = \frac{10 + 19}{2} = 14.5 \text{ minutes} = \frac{29}{120} \text{ hour.}$$

The rates differ per type of patient:

- for emergency patients $\lambda_E = 24$ per day = 1 per hour.
- 21 inpatients arrive during office hours and 6 outside office hours. So, in total $\lambda_I = 27$ per day = 1.125 per hour.
- outpatients only arrive during weekdays. On those weekdays 23 requests arrive. Outpatients show up with a probability of 84%, that means $23 \cdot 0.84 = 19.32$ outpatients arrive on weekdays. In the long-run we also need to account for the weekends. In the weekend, no outpatients arrive. Hence the average number of outpatients that arrive is $\lambda_O = 19.32 \cdot \frac{5}{7} = 13.8$ per day = 0.575 per hour.

So, the total rate of patients which arrive at the CT-department is $\lambda = \lambda_E + \lambda_I + \lambda_O = 1 + 1.125 + 0.575 = 2.7$ per hour.

We can define the workload (W) for a scanner as the rate of incoming patients multiplied by the scan duration, that is

$$W = \lambda \cdot E[\text{scan}] = 2.7 \cdot \frac{29}{120} = 0.6525 \text{ hours per hour} = 15.66 \text{ hours per day.}$$

To determine the average scan capacity we note that during office hours (08:00-16:00) two scanners are turned on and outside office hours (16:00-08:00) only one scanner is turned on. That means the total capacity C in the long-run is

$$C = 2 \cdot 8 + 1 \cdot 16 = 32 \text{ hours per day.}$$

So, the total utilization can be calculated as

$$\rho = \frac{W}{C} = \frac{15.66}{32} \approx 0.49 < 1.$$

Since the utilization is strictly smaller than 1, we can conclude that using two scanners during office hours and one outside offers is sufficient to be able to treat all patients.

3.2 Assumptions

To be able to make a simulation we need to make some assumptions. These are as follows:

- We scan every patient that arrives as soon as possible. That means that we assign the first possible slot that is available within the restrictions of the model to a patient that arrives.
- There are three types of patients (emergency, inpatient, outpatient). For each patient the following information is known: their type is fixed, the request time is known (outpatients), the arrival time in waiting room is known, outpatients have a fixed appointment time and are not too late, the time when the scan starts is known, and the scan duration is uniformly distributed on [10, 19].
- There are two CT scanners which work identical. Therefore we can define the capacity as:

$$S_i = \begin{cases} 2, & \text{office hours} \\ 1, & \text{else.} \end{cases}$$

When the office hours are over, we will always shut off one of the scanners. When both are still busy, the first that is finished is turned off.

- Emergency patients are always prioritized (non-preemptive). All the other patients wait in a FIFO queue.
- The waiting room consists of 3 chairs. If all chairs are occupied, patients will wait outside the waiting room. This has no effect on the arrival of patients.
- We assume there are no holidays or other exceptions that change the schedule of office hours. Therefore the office hours will always be during weekdays from 08:00 to 16:00.
- If all chairs in the waiting room are occupied and a patient waits outside, it stays outside. Also when a chair becomes available.
- Outpatients can only make their appointment on full hours.

3.3 Arrival of patients

The three types of patients all have their own arrival process.

3.3.1 Emergency patients

Emergency patients arrive according to a Poisson process with rate $\lambda_E = 24$ per day. This does not differ outside office hours.

3.3.2 Inpatients

Inpatient scan requests follow a Poisson process. The baseline arrival rate is denoted by λ_I , with approximately 21 requests arriving during office hours and 6 outside office hours on a weekday. During office hours, the arrival rate changes over time. It is constant before 09:00 and after 15:00, while between 09:00 and 15:00 it differs with a periodic pattern with two peaks: one at 10:30 and another at 13:30. This we can model as the following sine function:

$$\lambda_I(t) = \begin{cases} \frac{3}{8} + 3 \sin^2\left(\frac{\pi}{3}(t-9)\right), & \text{if } t \in [9, 15] \\ \frac{3}{8}, & \text{if } t \in [0, 9), (15, 24] \end{cases} . \quad (1)$$

When an inpatient request arrives, the patient remains on the ward until the CT department requests transport. The CT department has at most one inpatient in the waiting room. If no inpatient is present in the system, the ward is contacted immediately. Otherwise, the next inpatient is requested as soon as the current inpatient enters the scanning process. After a request, the inpatient arrival time at the CT department is uniformly distributed on [9, 15] minutes.

3.3.3 Outpatients

Outpatients call with a Poisson process with rate $\lambda = 23$ per weekday. Then an appointment is scheduled the earliest the next day. For outpatients there are only slots available on weekdays. 4 per hour in the morning, and 3 per hour in the afternoon. If there is no space in the same week anymore, patients are placed on a waiting list. On Friday all patients on the waiting list are scheduled in the next week. Outpatients that are scheduled have a probability of $p = 0.84$ of showing up.

3.4 State variables

To be able to determine performance measures using the simulation, we define the following state variables:

- A scanner can be either in state busy or idle. We define a capacity of either 2 scanners during office hours and 1 scanner outside office hours.
- There is a FIFO queue of waiting patients in the waiting room (or outside).
- The number of seats taken in the waiting room (≤ 3).
- A calendar with a number of appointments of outpatients on a certain day, at a certain hour.
- A waiting list of outpatients that wait over the weekend to have an appointment scheduled.
- In indication of whether we are in office hours or not.
- The slot capacity at a certain time.
- For patients we keep track of their type, time of arrival, service time, request time and whether their request came in during office hours.
- Whether there is an inpatient in the waiting room of the CT-department or in transport to the CT-department.
- The number of pending inpatient requests.

3.5 Events

To create a working simulation, we define the following events.

3.5.1 Completion of a scan

The scan completion event is triggered every time a scan of a patient is completed. When this happens, we update the time and utilization (via the `update_time` function). Furthermore we set the scanner to idle and try to start a new scan (by calling the `try_start_scan` function). A flowchart of this event can be found in figure 1.

3.5.2 Arrival of an emergency patient

The event is triggered when an emergency patient arrives. It then first updates time (utilization). It places the patient in the queue and determines whether there is still space in the waiting room (using the `enter_waiting_room` function). After that, it tries to start a scan and schedules a new arrival of an emergency patient. A flowchart of this event can be found in figure 2.

3.5.3 Arrival of an outpatient request

When an outpatient calls for an appointment, first all times are updated (utilization). We then try to schedule the outpatient in that week using `schedule_outpatient`. There it is either scheduled in this week and an arrival is planned, or placed on a waiting list, to be scheduled on Friday evening. After that the next call of an outpatient is scheduled. A flowchart of this event can be found in figure 3.

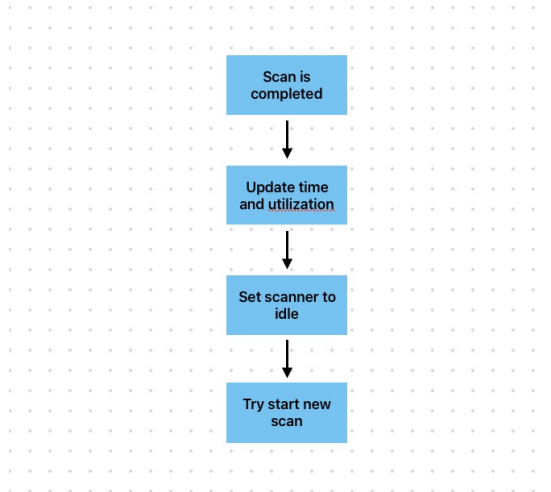


Figure 1: Flowchart for the scan completion event.

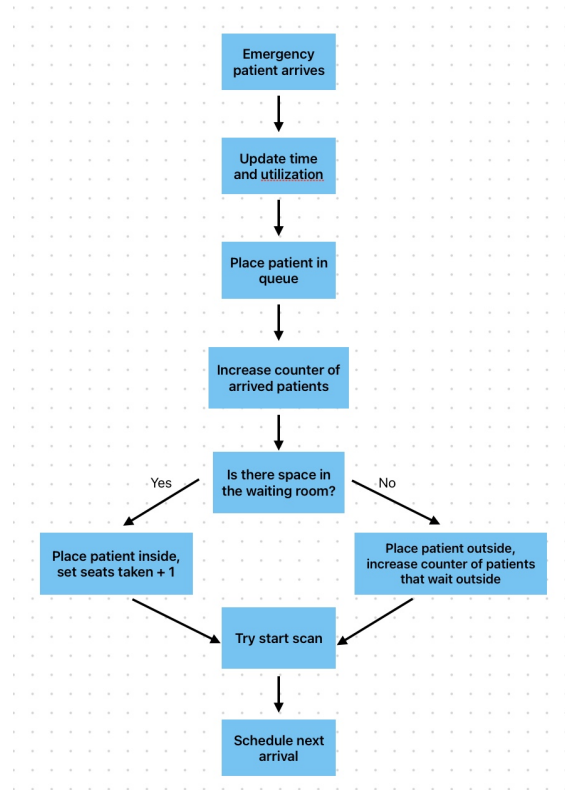


Figure 2: Flowchart for the event of an arrival of an emergency patient.

3.5.4 Weekly flush of the waiting list

The weekly flush is triggered every week on Friday at 16:00. Again, we first update time. After that we call `flush_waiting_list`, that schedules the requests on the waiting list in the next week (if possible). Then it determines the next Friday and schedules a weekly flush event at 16:00 on that day. A flowchart of this event can be found in figure 4.

3.5.5 Arrival of an inpatient request

This event occurs when a request for an inpatient is made from another ward in the hospital. It first updates the time and increases the number of pending inpatients. If there is no inpatient currently in the CT-department or in transport, then the inpatient goes in transport to the CT-department and an arrival is scheduled. Lastly, also a new arrival of a request is scheduled. A flowchart of this event can be found in figure 5.

3.5.6 Arrival of an inpatient in the CT-department

When an inpatient arrives in the CT-department, first the time is updated again. Also, the inpatient is removed from transport and placed in the waiting room. It is also checked whether there is space in the waiting room, or whether the inpatient should wait outside. Next to that, there will be determined whether the request was

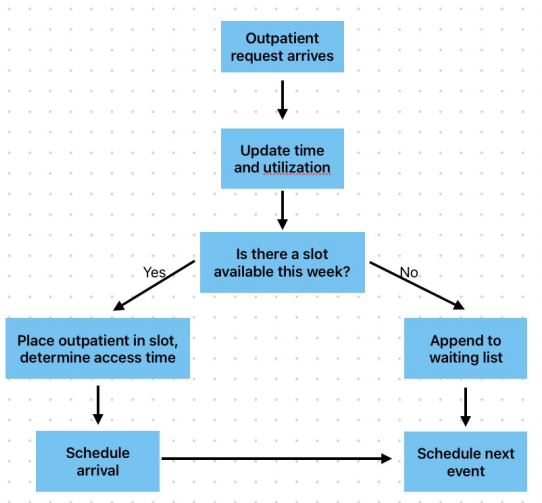


Figure 3: Flowchart for the outpatient request call event.

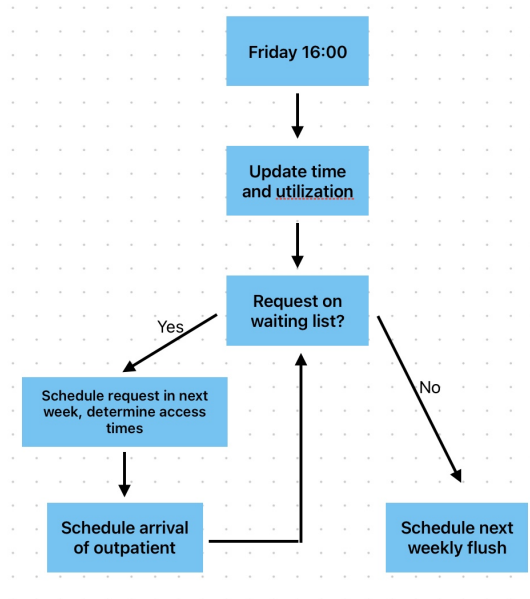


Figure 4: Flowchart for the weekly flush event.

made during office hours. If so, then a counter is increased of the amount of inpatient requests within office hours. At last, we try to start a scan. A flowchart of this event can be found in figure 6.

3.5.7 Arrival of an outpatient in the CT-department

Upon a trigger of this event, we first update time. Since outpatients have a probability of showing up of 84% we first determine whether the patient actually shows up. If so, then we will let it enter the queue, where again is determined if there is enough space in the waiting room. After that, we also try to start a scan. A flowchart of this event can be found in figure 7.

3.5.8 Try to start a new scan

Trying to start is actually not an event. But since this function is quite important we will give a description as well. Try to start a new scan is called by every event. We first determine how many scanners are active, for the active scanners we check whether they are idle or busy. In the case of an idle scanner we get the next patient from the queue (if there is any) by using `get_next_patient`. Which determines if there is an emergency patient waiting, and otherwise gives the first in- or outpatient. Then depending on the type of patient we store the waiting time or determine if a request that was made during office hours is also finished within office hours. For inpatients we also decrease the number of pending inpatients requests, and if there is another pending request, then we place that in transport. Lastly, we schedule the scan completion. A flowchart of this event can be found in figure 8.

3.6 Verification of simulation model

To verify the simulation model we give a description of justifying the implemented code, and we show some verification cases.

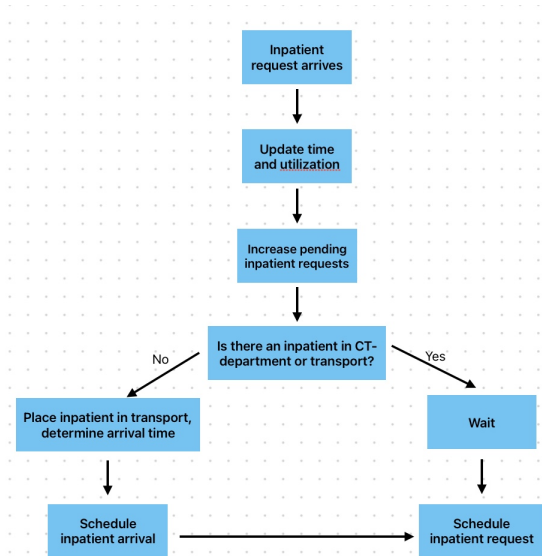


Figure 5: Flowchart for the event of an inpatient request.

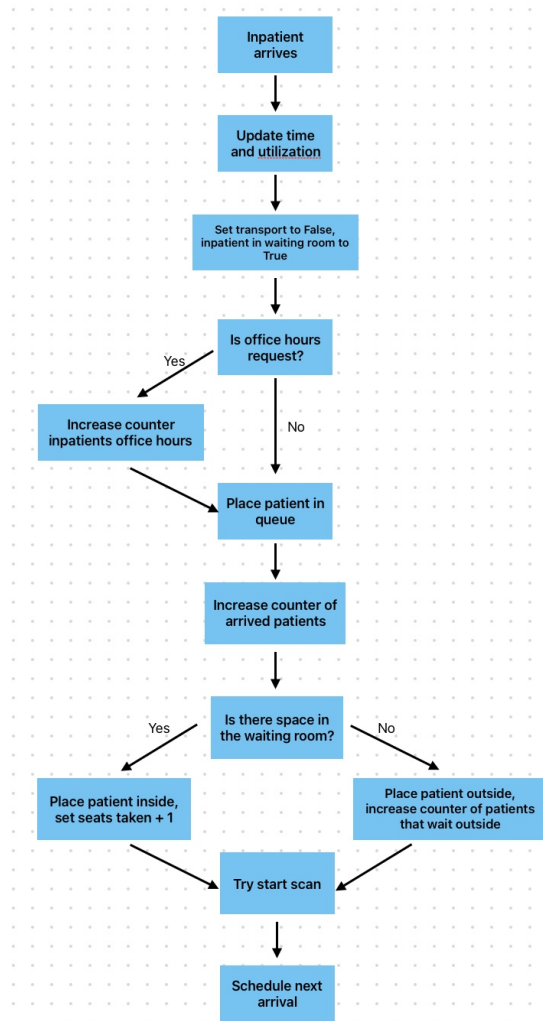


Figure 6: Flowchart for the inpatient arrival event.

3.6.1 Description

- The emergency waiting time is computed as the difference between the moment an emergency patient arrives and the moment their scan begins. This is correct because the simulation records the arrival time when the patient enters the waiting room and start time when the scanner becomes available and the patient is selected. Since emergency patients always have priority in the queue, the measured waiting time reflects the true delay caused only by scanner availability.
- Outpatient waiting time is measured as the time between the scheduled appointment arrival and the actual scan start. This is correct because outpatients are inserted into the FIFO queue with a known arrival time equal to their appointment time, and the simulation records a start time when a scanner becomes free. The difference between start time and arrival time therefore captures exactly the waiting room delay, independent of the scheduling.

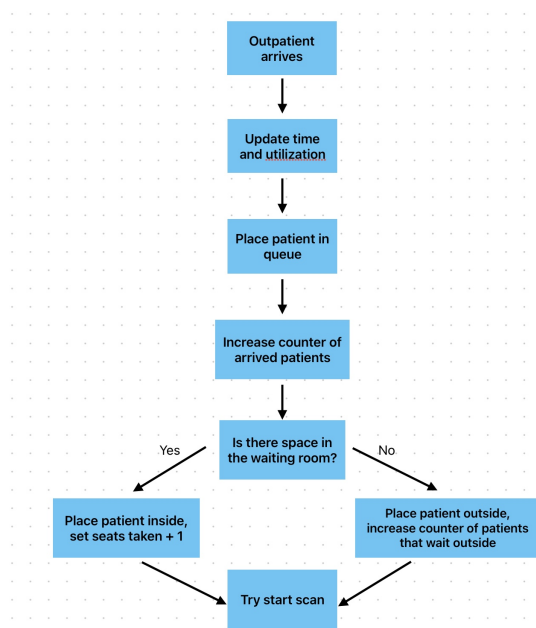


Figure 7: Flowchart for the outpatient arrival event.

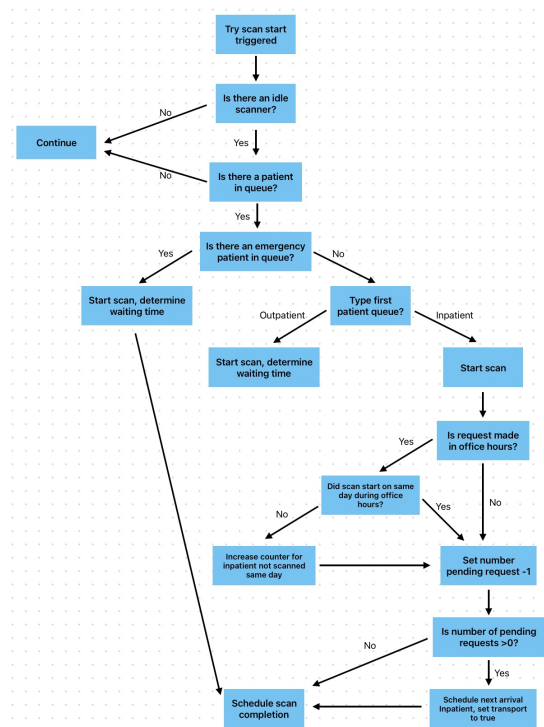


Figure 8: Flowchart for the try start scan procedure.

- The fraction of patients who must wait outside is computed by counting how many patients enter the waiting room when all seats are occupied and dividing by the total number of arrivals. This is correct because the simulation updates the number of seats that are occupied whenever a patient enters or leaves the waiting room, and increments the number of patients that need to wait outside of the waiting room only when all seats are occupied.
- The metric of inpatients with requests during office hours counts inpatients whose request arrives during office hours but whose scan begins either after 16:00 or on a later day. The simulation checks the request day and the scan start day and increments the counter only when they differ or when the scan starts after office hours.
- Office-hours utilization is computed by integrating busy time only over intervals where we are in office hours. Because the simulation breaks each event interval at the exact moment office hours begin or end, it is ensured that the computed utilization is only the period within office hours.
- Outside office hours utilization is computed analogously, integrating busy time only when we are outside office hours.
- Outpatient access time is computed as the number of days between the moment an outpatient calls and the moment of their scheduled appointment. This is correct because the simulation records the request time when the call arrives and computes the appointment time when a slot is assigned. The difference, divided by 24, yields the exact access delay in days (when rounded down).

3.6.2 Only arrival of emergency patients with exponential service time (M/M/c queue)

If we would only consider the arrival of emergency patients. Then the arrival process is a Poisson process. If we change the service time (=scan time) to a exponentially distributed random variable, then we obtain a M/M/c queue. For such a queue we know $\rho = \frac{\lambda}{c\mu}$, where λ is the arrival rate, c the number of servers, and μ the service rate. So, for one server outside office hours we would have $\rho = \frac{1}{1.4} = 0.25$, and for 2 servers during office hours $\rho = \frac{1}{2.4} = 0.125$.

We can also fix the number of scanners to two to obtain an explicit M/M/2 queue. For that we can also verify the waiting time. For the M/M/2 queue, we know $W_q = \frac{L_q}{\lambda}$, $L_q = \frac{(\lambda/\mu)^c \rho}{c!(1-\rho)^2} P_0$, $P_0 = \left(\frac{(\lambda/\mu)^0}{0!} + \frac{(\lambda/\mu)^1}{1!} + \frac{(\lambda/\mu)^c}{c!(1-\rho)} \right)^{-1}$. Specifically for $\lambda = 1$ and $\mu = 4$ we get:

$$P_0 = \left(1 + \frac{1/4}{1} + \frac{(1/4)^2}{2!(1-0.125)} \right)^{-1} = 0.7778 \quad (2)$$

$$L_q = \frac{(1/4)^2 \cdot 0.125}{2!(1-0.125)^2} \cdot 0.7778 = 0.00396 \quad (3)$$

$$W_q = \frac{0.00396}{1} = 0.00396 \quad (4)$$

3.6.3 Overload system

Another way to test the simulation model is to take a look at behavior in extreme cases. We can overload the system with patients per type of patients. Theoretically, we would then expect the following per type of patient:

- If we increase the number of emergency patients, we would expect that the utilization of the scanners converges to 1. Furthermore, we would expect that the emergency waiting time increases until capacity is reached and after that the waiting time would explode. Outpatients waiting times will increase too, but since emergency patients have priority they will not be scanned at all (so in the simulation the waiting time will be zero). The access time will not change, since this only depends on the number of scheduled outpatient for a certain hour on a certain day. The fraction of patients waiting outside the waiting room will converge to 1 since there will be a lot of patients in the waiting room after some time. Lastly, the inpatients scanned within the same day will drop to zero. This because emergency patients have priority over inpatients. Therefore, inpatients will not be scanned at all anymore.
- If we increase the number of inpatients, we would still expect that the utilization goes to 1 and the access time for outpatients stays the same by the same reasoning as before. However, in this case the emergency waiting time should not increase, because these patients are prioritized. Outpatient waiting time will increase, because they are still scanned in this case. Only way less outpatients will be scanned in a certain time and therefore the waiting time will increase. However, the fraction of patients waiting outside of the waiting room shall not increase much. Inpatients are only called to the waiting room when the previous patient is being scanned.
- If we would increase the rate of outpatients that call for an appointment, we would expect that everything stays the same or only increase lightly, except for the access time of outpatients. This is because there will be more calls. Hence, more outpatients need to be scheduled, but we cannot schedule more outpatients than the restrictions. So other metrics will not change.

3.6.4 Empty system

Another extreme case is when we would almost have an empty system. If we set the rates to zero, we would expect zero utilization, zero waiting times, the fraction of patients waiting outside the be zero, zero access times and the percentage of inpatients not being scanned on the same day to be zero.

4 Statistical Analysis

We examine the system using Discrete Event Simulation, where we provide 95% confidence intervals for the following performance indicators with 10% relative precision.

- CT scanner utilization, with a distinction between office hours and outside office hours.
- Access times for outpatients (the number of days between the request and the actual appointment).
- Waiting times for emergency patients and for outpatients (the time between the patient's arrival in the waiting room and the start of the scan).
- The fraction of (CT) patients who have to wait outside the waiting room.
- For inpatients with requests during office hours, the percentage of those that cannot be scanned before 16:00 on the same weekday.

To be able to calculate these metrics, we keep track of the following statistical counters during the simulation:

- The number of patients in the waiting room.
- The sum of the waiting times of emergency patients.
- The number of emergency patients that started their scan.
- The sum of the waiting times of outpatients.
- The number of outpatients that started their scan.
- The number of patients that arrive in the waiting room.
- The number of patients that have to waiting outside of the waiting room.
- The number of inpatients that arrive during office hours.
- The number of inpatients that arrived during office hours, but that could not be scanned that same day during office hours.
- The time the scanners are busy during office hours.
- The time scanners are busy outside office hours.
- The time scanners are turned on during office hours.
- The time scanners are turned on outside office hours.
- The sum of access times of outpatients.

- The number of outpatients that make an appointment.

We make use of the batch means method. We keep track of these variables per batch. We also define the time of the last event, the warmup time, the batch start time, the batch length and the current batch as helpers. At the end of the batch we compute the asked metrics for the given batch and store these. We compute the metrics as:

- Scanner utilization during office hours = the time the scanners are busy during office hours / the time scanners are turned on during office hours.
- Scanner utilization outside office hours = the time the scanners are busy outside office hours / the time scanners are turned on outside office hours.
- Access times for outpatients = the sum of access times of outpatients / the number of outpatients that make an appointment.
- Waiting times for emergency patients = the sum of the waiting times of emergency patients / the number of emergency patients that started their scan.
- Waiting times for outpatients = the sum of the waiting times of outpatients / the number of outpatients that started their scan.
- the fraction of patients who have to wait outside the waiting room = the number of patients waiting outside of the waiting room / the number of patients that arrive in the waiting room.
- The percentage of inpatients that cannot be scanned before 16:00 on the same weekday = The number of inpatients that arrived during office hours, but that could not be scanned that same day during office hours / the number of inpatients that arrive during office hours.

At the end of a batch we store these in a list and then reset the variables. After a number of batches we use this results to compute the metric in steady state and the 95%-confidence intervals.

5 Results

In this section we will show the results for both the verification cases, as well as the general case.

5.1 Verification cases

5.1.1 M/M/c system

If we run the simulation with only the arrival of emergency patients, we obtain the values for the utilization of scanners during office hours and outside office hours as shown in table 1

Parameter	Value	95% CI
Utilization outside office hours	0.2480	[0.2468, 0.2492]
Utilization during office hours	0.1253	[0.1242, 0.1264]

Table 1: Utilization of scanners for reduction to a M/M/c queue

We see that these values and confidence intervals match the theoretical values for the utilization.

If we also fix the number of scanners to 2, we obtain the values for the waiting time of emergency patients as shown in table 2.

Parameter	Value	95% CI
Emergency waiting time	0.0039485	[0.0038886, 0.0040083]

Table 2: Waiting time of emergency patients for reduction to a M/M/2 queue.

5.1.2 Overload

If we run the simulation with rate $\lambda_E = 10$ (letting the other rates stay the same), we get the results presented in table 3. Since more patients are arriving, the running of batches takes longer. Therefore we chose to run a maximum of fifty batches. The relative precision we obtained with that is 0.1601.

Parameter	Value	95%CI
Emergency Waiting Time	95363	[80092, 110634]
Outpatient Waiting Time	0.0*	[0.0, 0.0]
Fraction Waiting Outside	0.999998	[0.999996, 0.9999998]
Percentage Inpatient Not Same Day	0.0*	[0.0, 0.0]
CT Utilization (office hours)	1.0	[1.0, 1.0]
CT Utilization (outside office hours)	1.00485	[1.00480, 1.00489]
Outpatient access time (days)	1.37742	[1.37271, 1.38214]

Table 3: Values and 95%-confidence intervals for $\lambda_E = 10$. The * indicates that no patient is scanned and therefore the value is 0.0.

We see that these results match our expectations.

If we run the simulation with fixed rate $\lambda_I = 60$ (and the target precision of 0.05 is reached) we get the results presented in table 4.

We see that the emergency and outpatient waiting time increase, as well as the fraction of patients that wait outside and the utilization for both during office hours and outside office hours. This is because more patients arrive, thus the utilization is higher, thus the waiting times increase. Even for emergency patients, because they see the busy scanner upon arrival more often.

Parameter	Value	95%CI
Emergency Waiting Time	0.13628	[0.13402, 0.13854]
Outpatient Waiting Time	0.19766	[0.19400, 0.20133]
Fraction Waiting Outside	0.02356	[0.02240, 0.02473]
Percentage Inpatient Not Same Day	99.998	[99.982, 100.014]
CT Utilization (office hours)	0.83843	[0.83496, 0.84187]
CT Utilization (outside office hours)	0.98580	[0.98526, 0.98633]
Outpatient access time (days)	1.3894	[1.36134, 1.41745]

Table 4: Values and 95%-confidence intervals for $\lambda_I = 60$.

If we run the simulation with a rate of $\lambda_O = 50$ and keep the rest the same (relative precision 0.1620). We get the results in table 5.

We see that these results again meet our exceptions. We notice a slight increase of the percentage of inpatients that cannot be scanned on the same day, as well as the CT utilization during office hours and the

fraction of patients that have to wait outside. We can explain this by the fact that there are more outpatients scheduled (within restrictions), and therefore scanners are a bit busier.

Parameter	Value	95%CI
Emergency Waiting Time	0.05183	[0.05065, 0.05301]
Outpatient Waiting Time	0.13378	[0.13183, 0.13572]
Fraction Waiting Outside	0.01575	[0.01477, 0.01673]
Percentage Inpatient Not Same Day	1.26667	[1.06151, 1.47288]
CT Utilization (office hours)	0.65740	[0.65353, 0.66126]
CT Utilization (outside office hours)	0.33248	[0.32958, 0.33538]
Outpatient access time (days)	932.45	[858.26, 1006.64]

Table 5: Values and 95%-confidence intervals for $\lambda_O = 10$.

5.1.3 Zero system

For the zero system verification case we set $\lambda_O = \lambda_I = 0$ and $\lambda_E = 0.01$, because the simulation will break when no patients arrive. We obtain the results presented in table 6. These results match the expected outcome.

Parameter	Value	95%CI
Emergency Waiting Time	0.00024	[0.00023, 0.00025]
Outpatient Waiting Time	0.0	[0.0, 0.0]
Fraction Waiting Outside	0.0	[0.0, 0.0]
Percentage Inpatient Not Same Day	0.0	[0.0, 0.0]
CT Utilization (office hours)	0.00121	[0.00121, 0.00121]
CT Utilization (outside office hours)	0.00242	[0.00241, 0.00242]
Outpatient access time (days)	0.0	[0.0, 0.0]

Table 6: Values and 95%-confidence intervals for $\lambda_I = \lambda_O = 0, \lambda_E = 0.01$.

5.2 General case

Doing the simulation with the given parameters, we obtain the results shown in table 7

Parameter	Value	95%CI
Emergency Waiting Time	0.05087	[0.05056, 0.05118]
Outpatient Waiting Time	0.13647	[0.13598, 0.13696]
Fraction Waiting Outside	0.01452	[0.01431, 0.01474]
Percentage Inpatient Not Same Day	0.84529	[0.80303, 0.88755]
CT Utilization (office hours)	0.59509	[0.59389, 0.59630]
CT Utilization (outside office hours)	0.33149	[0.33076, 0.33223]
Outpatient access time (days)	1.37638	[1.37329, 1.37947]

Table 7: Results of the simulation with the given parameters.

6 Conclusion

This report examined whether the CT-scan department's current scheduling policy is able to handle patient demand sufficiently. Both the analytical capacity check and the simulation results show that the available scanner capacity is sufficient, the scanners are not overloaded.

Emergency patients, who require immediate attention, have a very short waiting time of around 3 minutes. Outpatients receive appointments within a reasonable number of days, namely 1.4 days, and their waiting time at the CT department remains limited to around 8 minutes. Most inpatients whose requests arrive during office hours are scanned on the same day, which is over 99%. The waiting room, despite its limited size, rarely reaches full occupancy (1.5%).

Overall, the current policy performs well. The department uses the scanners efficiently while ensuring timely care for all patient groups.